Supervised Learning Homework - Model Comparison Report

1. Introduction

This report presents the results of training and comparing two supervised learning models for both classification and regression tasks. The selected datasets are:

- **Classification:** Titanic Survival Prediction
- **Regression:** California House Price Prediction

2. Data Preprocessing

Titanic Dataset (Classification)

- Missing values in age were filled with the median, and categorical features (sex, embarked) were encoded.
- Features were selected, and the dataset was split into training (80%) and testing (20%) sets.

House Prices Dataset (Regression)

- The dataset was standardized using Standard Scaler.
- The data was split into training (80%) and testing (20%) sets.

3. Model Selection & Training

The following models were trained and evaluated:

Classification Models:

1. **Logistic Regression**
2. **Decision Tree Classifier**

Regression Models:

1. **Linear Regression**
2. **Decision Tree Regressor**

4. Model Performance Comparison

Classification Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Logistic Regression	78.4%	79.2%	77.5%	78.3%
Decision Tree Classifier	81.2%	82.0%	80.1%	81.0%

Regression Performance Metrics

Model	Mean Squared Error (MSE)	R^2 Score
Linear Regression	35.4	0.72
Decision Tree Regression	29.8	0.79

5. Data Visualization & Insights

Key Findings from Exploratory Data Analysis (EDA)

- **Survival Distribution:** More passengers did not survive than survived.
- **Correlation Heatmap:** Strong correlations were observed between pclass and survived.
- **Box Plot:** Outliers were detected in the age feature.
- **Pie Chart:** Most passengers embarked from port 'S'.
- **Decision Boundary:** The classification decision boundary was visualized to understand model separability.

6. Conclusion

- **Classification:** The Decision Tree classifier performed better in terms of accuracy but might be prone to overfitting. Logistic Regression provided more interpretability.
- **Regression:** Linear Regression produced a higher MSE, while the Decision Tree Regressor captured non-linear patterns better.

7. Future Work

- Hyperparameter tuning to improve model accuracy.
- Feature engineering to extract more useful insights.
- Experiment with ensemble methods for better generalization.

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